Examiner's Use

5 **(a)** For a particular gas, the emission and absorption spectra are obtained for the visible light spectrum.

Discuss one similarity and one difference between the discrete lines of the absorption and emission spectra of this gas.

……

……

……

……

…… [2]

(b) Fig. 5.1 gives information on three lines observed in the emission spectrum of hydrogen atoms.

wavelength / nm	energy of photon / eV
486	2.56
656	
1880	0.66

Fig. 5.1

(i) Complete Fig. 5.1 by calculating the energy of the photon with wavelength 656 nm. [1]

(ii) Fig. 5.2 is a partially completed diagram to show energy levels of a hydrogen atom.

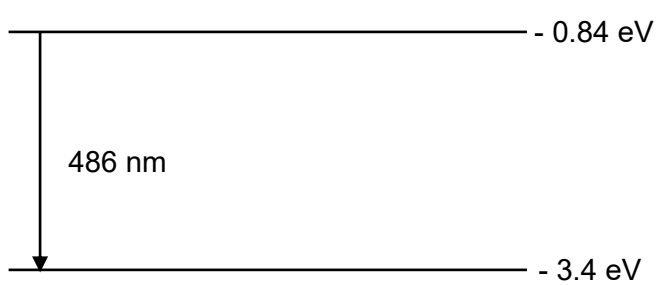

Fig. 5.2

On Fig. 5.2, draw an additional labelled energy level which will account for the emission of the photons in Fig. 5.1. [1]

For Examiner's Use

(c) Three of the energy levels of a lithium atom are shown in Fig. 5.3.

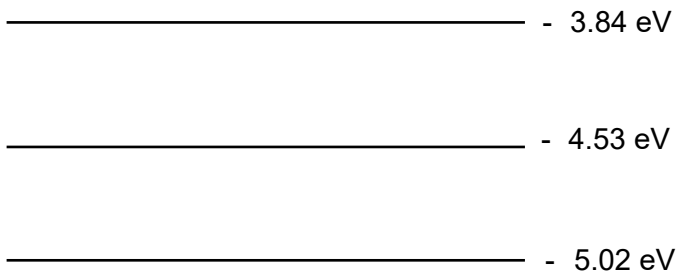

Fig. 5.3

One way to study the energy levels of an atom is to bombard the atom with electrons and measure the kinetic energies of the bombarding electrons before and after the collision. If a lithium atom which is originally in the – 5.02 eV level is bombarded with an electron of kinetic energy 0.92 eV, the scattered electron can have only two possible kinetic energies.

States these two kinetic energy values, and state what happens to the lithium atom in each case.

1st possible kinetic energy value: ……. eV [1]

……

…….. [1]

2nd possible kinetic energy value: ……. eV [1]

……

…….. [1]